

Gabrielle (Gaby) N. Altman

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PROFESSIONAL SUMMARY

Bioinformatics data scientist & PhD Candidate (defending October 1, 2026) with expertise in biobank-scale statistical human genomics. Proven track record in uncovering novel genotype-phenotype associations through 5+ years of research in tandem repeat expansions (TREs) and multiomic integration. Experienced in building scalable, reproducible pipelines within cloud and HPC environments to analyze 350,000+ genomes across the UK Biobank (UKB) and All of Us (AoU). Seeking industry R&D roles where I can apply this statistical genomics expertise to drive target discovery and human genetics-driven therapeutic development.

EDUCATION

Icahn School of Medicine at Mount Sinai (ISMMS)

PhD Candidate in Biomedical Sciences

Genetics and Genomic Sciences Multidisciplinary Training Area

Thesis defense date: October 1st, 2026

New York, NY

August 2020 – Present

Washington University in St. Louis (WashU)

Bachelor of Science in Biomedical Engineering (Minor: Computer Science)

Summa Cum Laude

St. Louis, MO

May 2020

GPA: 3.98/4.00

TECHNICAL SKILLS

Programming: R (tidyverse, ggplot2), Python, Bash/Unix, MATLAB

Statistical Genetics & Data Science: GWAS/PheWAS (REGENIE, PLINK, brglm), PRS, fine mapping (CAVIAR), conditional analysis, machine learning (scikit-learn, PyTorch)

Cloud: Google Cloud Platform (GCP), AWS, Terra, DNAnexus, All of Us Researcher Workbench, Cavatica

Scientific Computing: Docker, Git/GitHub, HPC (LSF, bjobs)

Genomics & Bioinformatics: WGS, RNA-seq, methylation arrays, long read sequencing, variant calling

Molecular biology: PCR, gel electrophoresis, bisulfite conversion, combined bisulfite restriction analysis

PROFESSIONAL & RESEARCH EXPERIENCE

Sharp Lab, Icahn School of Medicine at Mount Sinai

PhD Candidate

New York, NY

November 2020 – Present

- Discovered 19 novel tandem repeat (TR) expansion loci associated with neurodegenerative disease by meta-analyzing >350,000 samples (UKB and AoU) using REGENIE with approximate Firth logistic regression (manuscript in preparation).
- Engineered a composite neurodegenerative phenotype by combining multiple clinical conditions from EHR data into a single binary trait.
- Genotyped >77,000 TRs and identified expansions in >100,000 All of Us samples, accounting for batch effects and heterogeneity.
- Integrated WGS SNP, methylation array, bulk RNA-seq, and long-read sequencing across AMP PD, GTEx, TOPMed, ADSP, and AoU to characterize functional and clinical impact of TR expansions.
- Screened *LDLR* promoter hypermethylation in 308 mutation-negative familial hypercholesterolemia patients via Combined Bisulfite Restriction Analysis (manuscript under review).
- Developed containerized, reproducible cloud pipelines (Cavatica/AWS) to scale TR genotyping and *de novo* expansion calling for thousands of WGS trios in pediatric congenital disorders.

Brent & Doering Labs, Washington University in St. Louis

Undergraduate Researcher

St. Louis, MO

January 2018 – May 2020

- Analyzed genome sequences from Bulk Segregant Analysis (BSA) experiments using R to identify natural genetic variants of *Cryptococcus neoformans* that influence its virulence.
- Developed novel algorithms using R to simulate and optimize BSA experiments.
- Prepared crosses of *C. neoformans* and isolated their spores to create a library of recombinant strains.

Regeneron Pharmaceuticals, Inc., Inflammatory and Immune Diseases

Tarrytown, NY

R&D Intern

June 2017 – August 2017

- Analyzed serum IgE levels by conducting ELISA tests to evaluate *in vivo* mouse experiments on autoimmune diseases such as Asthma, Allergies, and SLE.
- Led a software validation experiment testing the use of Halo software to measure microscopic skin thickness in murine psoriasis models, reducing subjectivity in previous methods.

Selected other projects

- Gene Expression Extension (Ma'ayan Lab, 2021): Designed and built GEO2Appyter, a Google Chrome extension connecting Gene Expression Omnibus pages to an automated bulk RNA-seq pipeline.
- eQTL Analysis (Huckins Lab, 2020): Implemented an eQTL association analysis for "smoking-QTLs" and performed variance partitioning on brain gene expression data.
- WashU Biomedical Engineering Senior Capstone (2020): Awarded 1st place (out of 21) for designing a wearable post-operative monitor; led software development and circuitry.

PUBLICATIONS & PRESENTATIONS

Altman, G. N., Jadhav, B., Garg, P., Shadrina, M., Manigbas, C. A., Lee, W., Martin-Trujillo, A., & Sharp, A. J. Pleiotropic tandem repeat expansions in *DAPK1* are associated with neurodegenerative disorders. (Manuscript in preparation)

Altman, G. N., Martin-Trujillo, A., Hesson, L. B., Bell, D. A., Hooper, A. J., Sharp, A. J. Promoter methylation of *LDLR* is not a common cause of familial hypercholesterolemia. (Under revision)

Manigbas, C. A., Jadhav, B., Garg, P., Shadrina, M., Lee, W., **Altman, G.**, Martin-Trujillo, A., & Sharp, A. J. (2024). A phenome-wide association study of tandem repeat variation in 168,554 individuals from the UK Biobank. *Nature Communications*, 15(1), 10521. <https://doi.org/10.1038/s41467-024-54678-0>

Jadhav, B., et al., including **Altman, G.**, and Sharp, A. J. (2024). A phenome-wide association study of methylated GC-rich repeats identifies a GCC repeat expansion in *AFF3* associated with intellectual disability. *Nature Genetics*, 56(11), 2322-2332. <https://doi.org/10.1038/s41588-024-01917-1>

Altman G.N., Lee W., Jadhav B., Martin-Trujillo A., Sharp A.J., Analysis of pathogenic tandem repeat variation in congenital disorders (Abstract #2389). Poster presented at the 72nd Annual Meeting of The American Society of Human Genetics, October 2022, in Los Angeles, California.

AWARDS, TEACHING & LEADERSHIP

- Teaching Assistant: Practical Applications of Personal Genomes (ISMMS, 2023); Biomedical Sciences (ISMMS, 2021, 2022); Physiological Control Systems (WashU, 2019, 2020), Computer Science I (WashU, 2018)
- Select Honors: 1st Place in WashU BME Day design competition, Summa Cum Laude, Outstanding Senior & Junior Awards for Academic Excellence in BME at WashU (2019, 2020); Dean's List, Tau Beta Pi Engineering Honor Society
- Mentorship: PhD Peer Mentor (ISMMS, 2021-2024); mentored high schoolers in hands-on RNA-seq pipelines through the Center for Excellence in Youth Education (CEYE) (ISMMS, 2021)

PROFESSIONAL DEVELOPMENT

Advanced Gene Mapping Course

January 2022

The Rockefeller University

- Trained in end-to-end genomic discovery: from LMM-based association analysis (REGENIE) and data QC to functional annotation, PRS, and transcriptomic integration (TWAS).

62nd Annual McKusick Short Course in Human and Mammalian Genetics and Genomics

July 2021

The Jackson Laboratory (JAX)

- Completed an intensive two-week program covering advanced topics in mammalian genetics, high-throughput sequencing technologies, and statistical genomics.